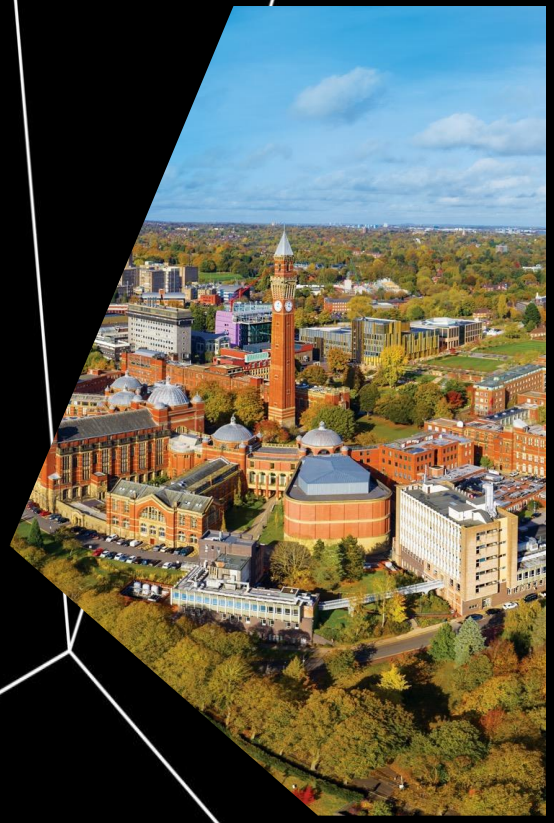
Module: FON

Somatosensory system

Lecture :8

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Objectives

- Primary sensory neurons and somatosensory system.
- Different receptors in skin and their properties
- Explain how are stimuli contacting the skin are transduced into neural activity. (**Partially explained here** and the **rest in the SGT** of **somatosensory system**)
- Pathways Carrying sensations of touch, pressure and vibration
- Pathways carrying sensations of pain and temperature to somatosensory cortex.
- Mapping of the somatosensory cortex

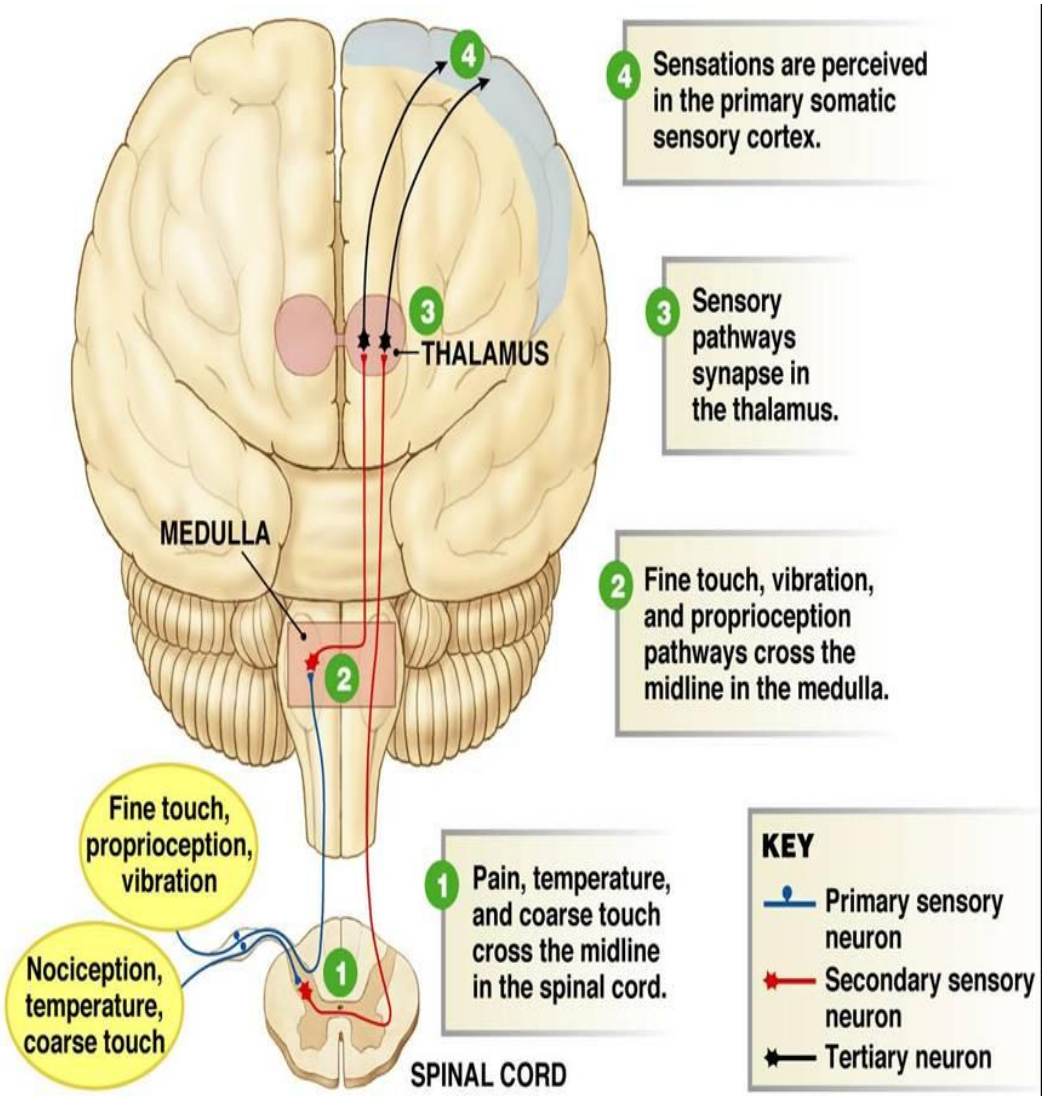
The somatosensory system

The neural sense concerned with body sensations; transmits information from sensory receptor organs to cerebral cortex

Origin: Greek Soma = body

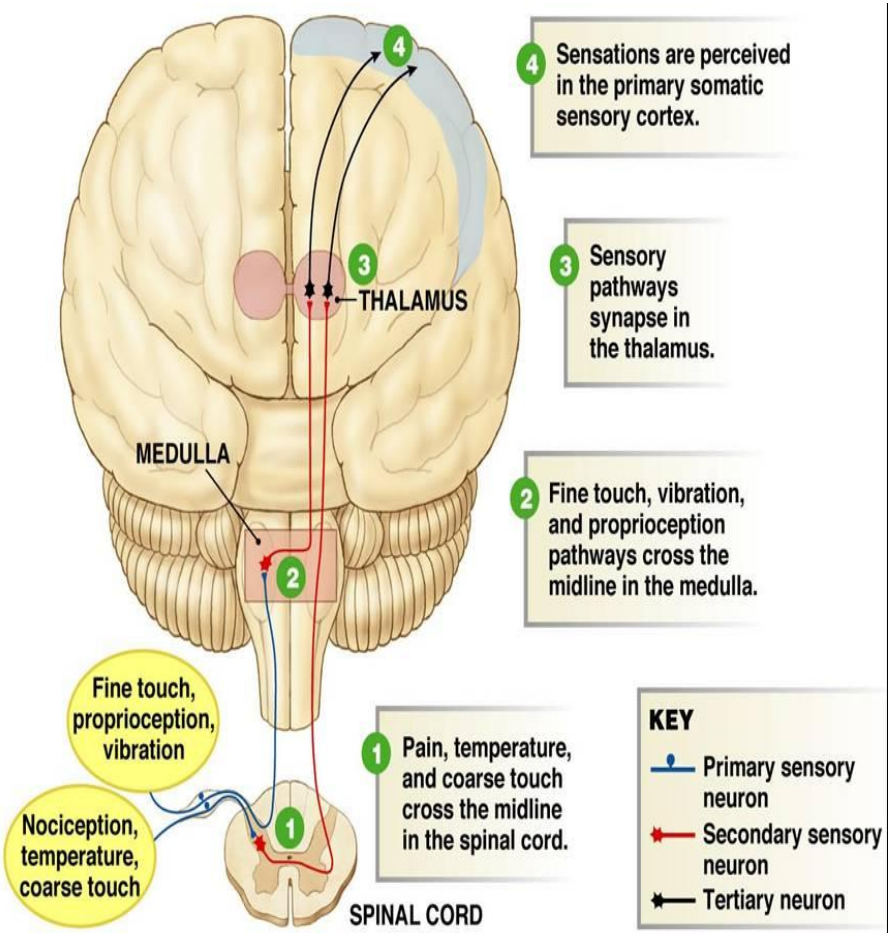
How Somatosensory system works (example)

- The somatosensory system transmits information from sensory receptor organs to cerebral cortex via series of neurons (Primary : Blue), Secondary : Red), Tertiary: Black neurons).



How Somatosensory system works (example)

- **Components** : carry sensations forn receptors of **skin, internal organs and deep tissues** and provide sensations, namely :
  - Tactile (Touch, pressure, vibration)
  - Somatic (Pain, temperature)
  - Proprioception (position of limbs and body in space).
  - Unconscious proprioception (Spinocerebellar pathway) [Not discussed]
  - Sensations from the face and head Trigeminothalamic pathways [Not discussed]

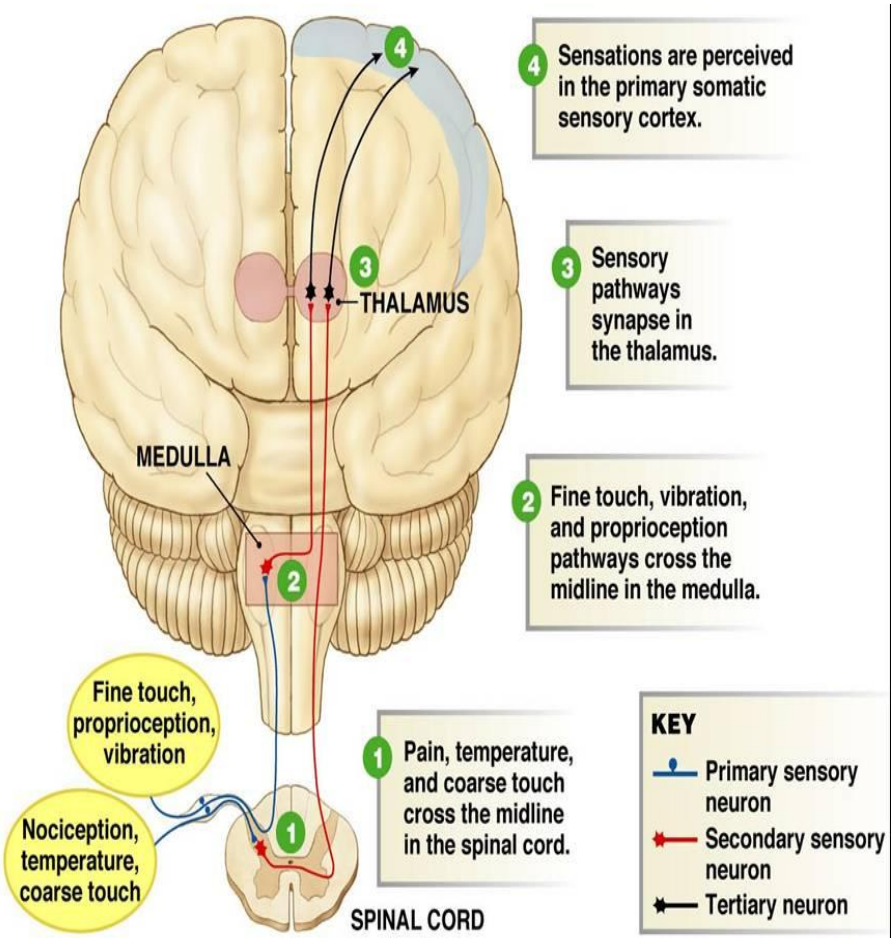


Note: Somatosensory: these are general senses (Somatic + Visceral)
Special senses: (Taste, Vision, Audition) processed via cranial nerves and has its own dedicated pathways.

How Somatosensory system works (example)

What does this system do (mainly):

- Helps perceiving a sensation (Touch, pressure, vibration etc).
- Maintain communication between body parts.
- Helps movement control ie; has it been executed properly or not by receiving signals from periphery.



What is **Sensation ?**

It is conscious / subconscious awareness of changes in external / internal ambience. Sensation nature & reaction type vary according to the ultimate destination. For eg: Sensory information to spinal cord act as input for spinal reflexes (stretch reflex) & when at brain stem level produce complex reflexes (eg: change of HR / Breathing rate)

What is **Perception ?**

Conscious awareness of a stimuli and interpretation of sensations like touch, pain, hearing etc. (Primarily a function of **cerebral cortex**)

What is **Stimulus ?**

Any sort of electrical, chemical, mechanical, physical, thermal agents/events that cause changes in external/ internal environment of a tissue to bring about a reaction/ response

What is **Receptor ?**

It detects information (Stimuli = Energy) about changes

What is **Coded Form/Coding in nervous system?**

Pattern of Action Potential

What does the code **represent ?**

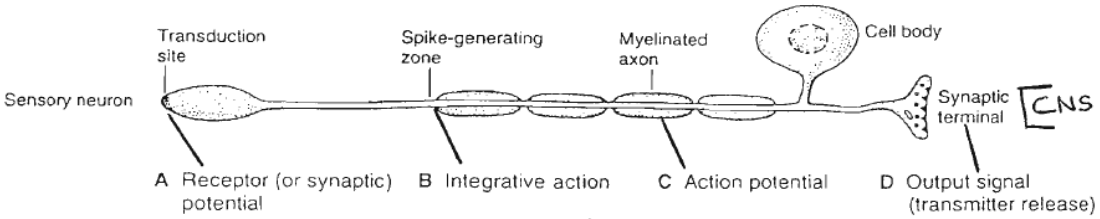
Modality : One feature of a stimulus. eg. Vision, Hearing, temperature - represented by different receptors by different pathways

Intensity : Represented by mean frequency of AP's

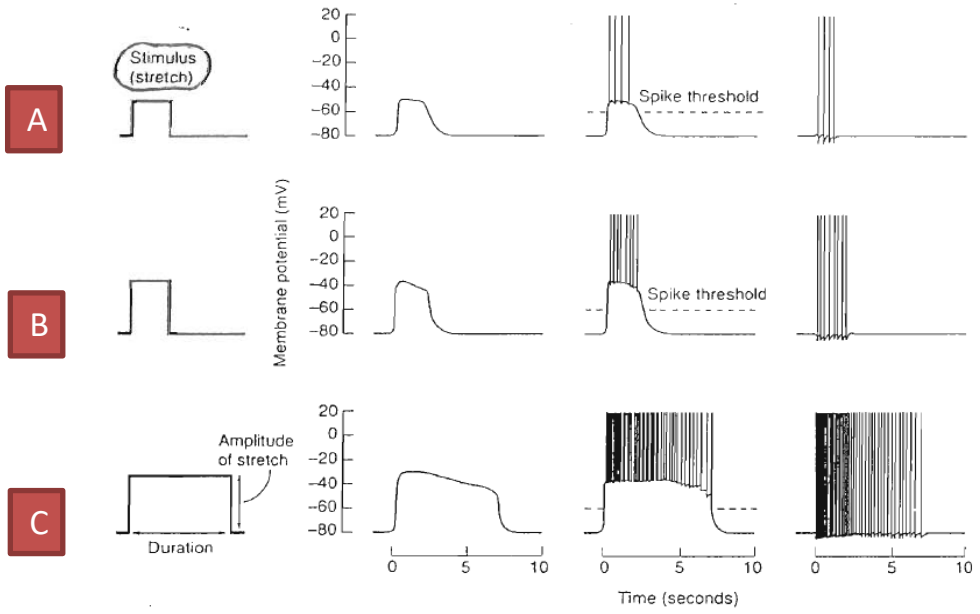
Duration: Represented by duration of AP's

Location: Represented by localized area (Receptive Field)

Sensory Coding (Example: Mechanoreceptors)



Coding is done generally to the extent the NT's are released form the nerve terminals in response to a stimulus.



A: Stretch leads few NT's release.

B: As stretch increases →firing levels of the neuron increases → more NT release (Brain sense stronger stimulus)

C: Prolonged stretch → extensive firing of the same neuron to send signal. (Brain sense longer stretch stimulus)

Variation of AP pattern is a common way of sensory coding

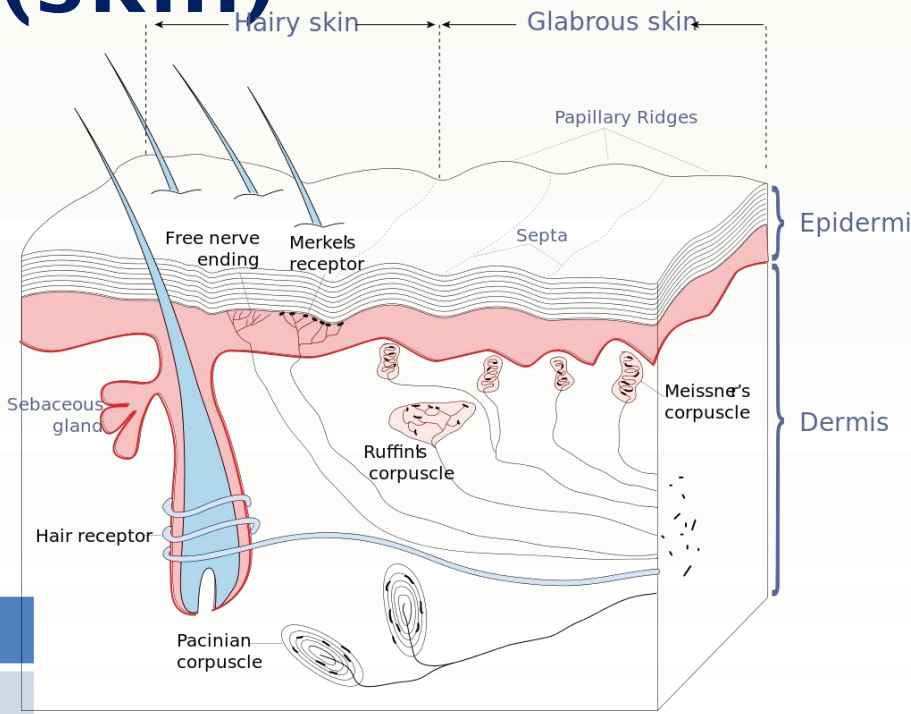
Sensory receptors

- Receptors can be classified based on the type of stimulus that activates them → mechanoreceptor, chemoreceptors, thermoreceptor, nociceptor, photoreceptor…

- **Mechanoreceptor**: mechanical force
- **Chemoreceptor**: chemicals
- **Nociceptor**: pressure, temperature, noxious chemicals

Sensory receptors (skin)

- Several special or sensory nerves can be found over skin (dermis and epidermis layer)



Specific receptors	Sensory modality
Pacinian corpuscle (**dermis**)	Vibration, pressure
Meissner's corpuscle (**dermis**)	Touch (two-point discrimination)
Ruffini's corpuscle (**dermis**)	Pressure (stretch)
Merkel's disc (**epidermis**)	Touch
Free nerve endings (**epidermis**)	Pain and temperature

Mechanoreceptors

Thermoreceptor & nociceptor

Four major types of mechanoreceptors

- **Merkel's receptor (disk)**
- **Meissner's corpuscle**

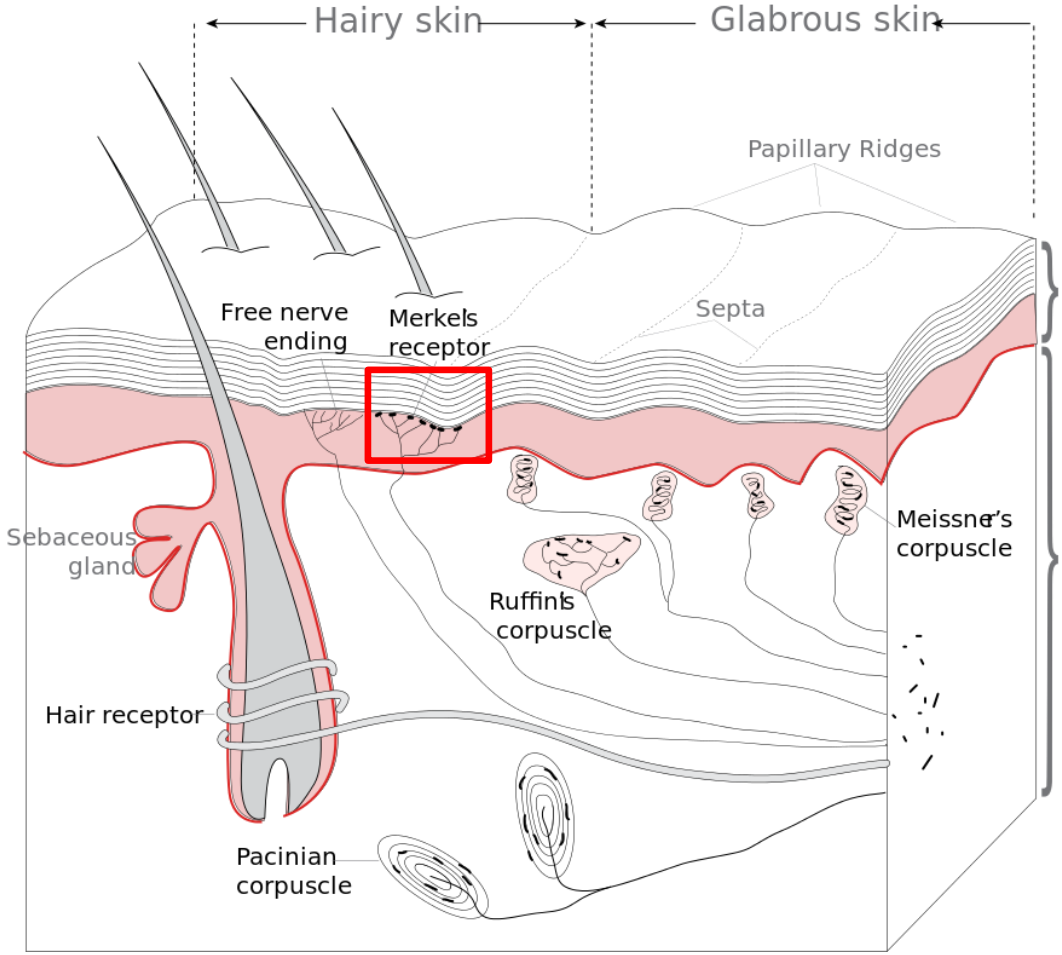
} more superficial layers of skin

- **Ruffini's corpuscle (ending)**
- **Pacinian corpuscle**

} deeper layers of skin

Note: Free nerve endings will be discussed in the "**Somatosensory SGT**" session

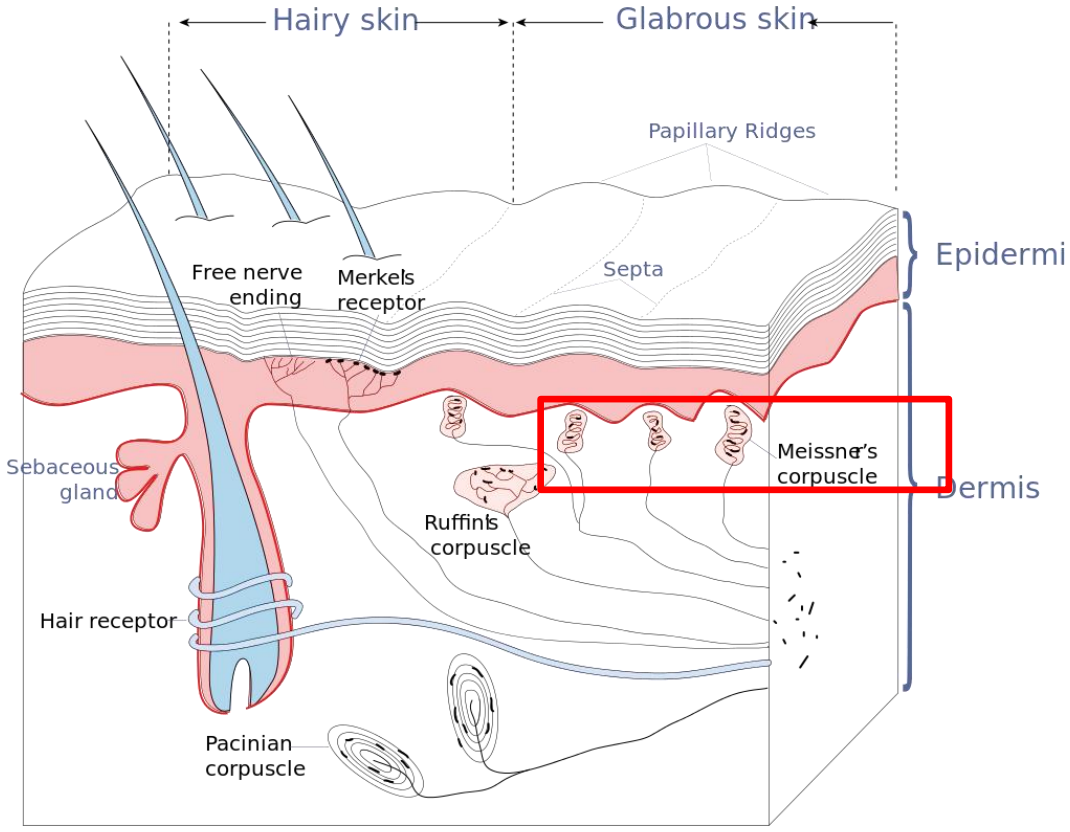
Merkel's receptor (disk) / Merkel cell–neurite complex



- high density in epidermis of digits and around mouth (50 per mm^2)
- lower density elsewhere on glabrous skin except fingertips
- very low density in hairy skin
- slowly adapting
- sustained light touch
- perception of form and texture

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Meissner's (or Tactile) corpuscles



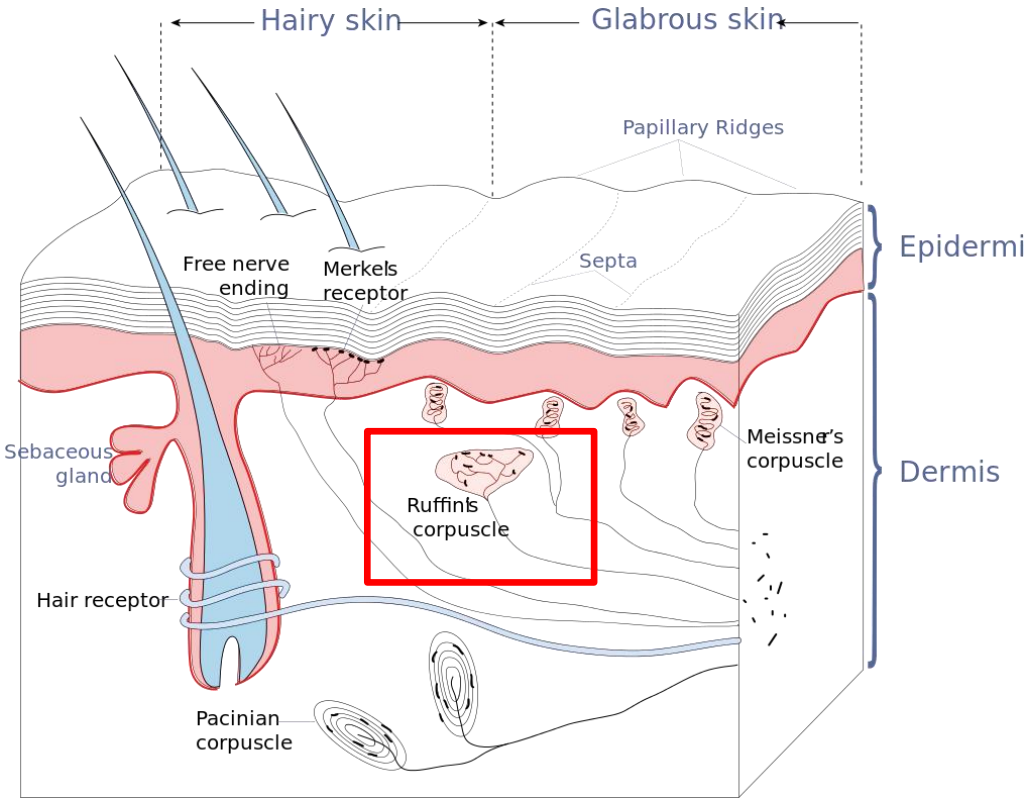
- rapidly adapting
- constantly changing stimulus required

e.g.

- detect putting on clothes, but not the wearing
- involved in adjustment of *grip force* when objects are lifted
- light touch (two-point discrimination)
- Vibration

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Ruffini's (or Bulbous) corpuscle

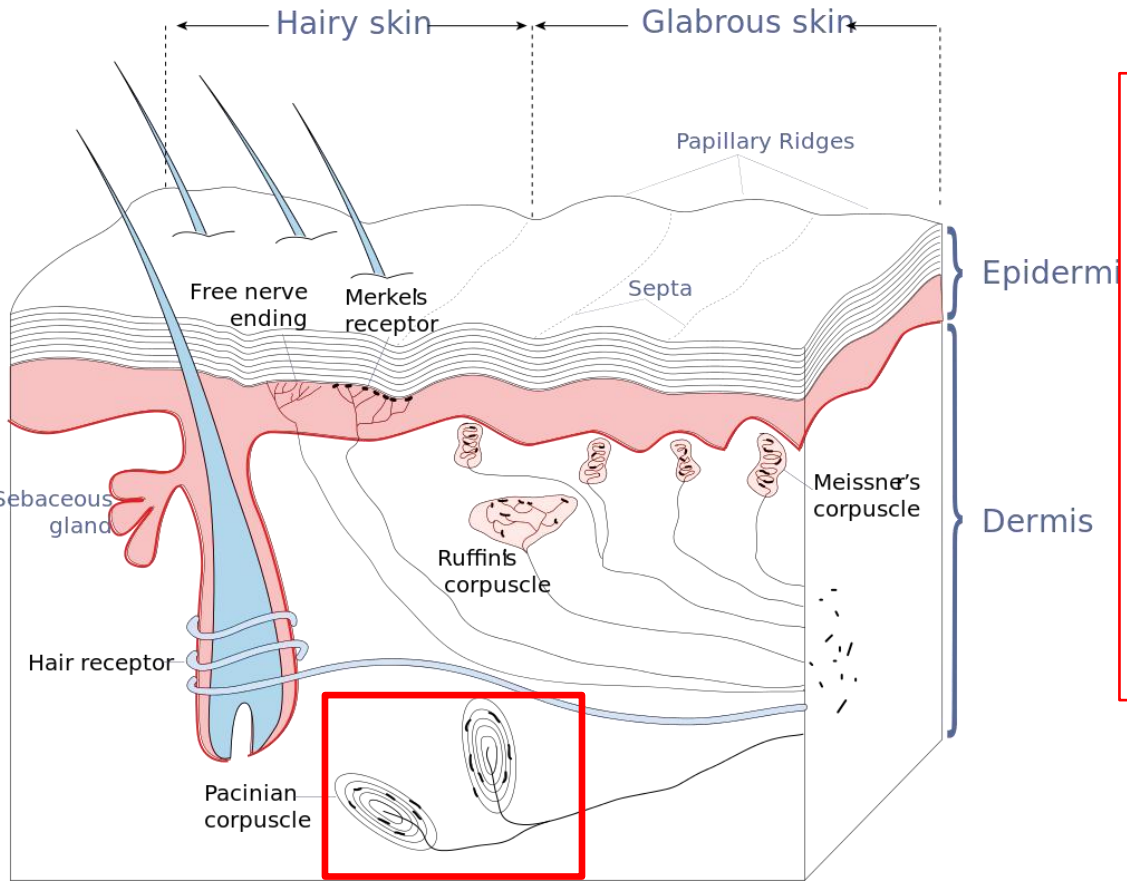


- respond to lateral movement or stretching of skin
- deep touch, involved in monitoring grasped object *slippage*
- *Slowly adapted*

⇒ *reflex adjustment of grip force*

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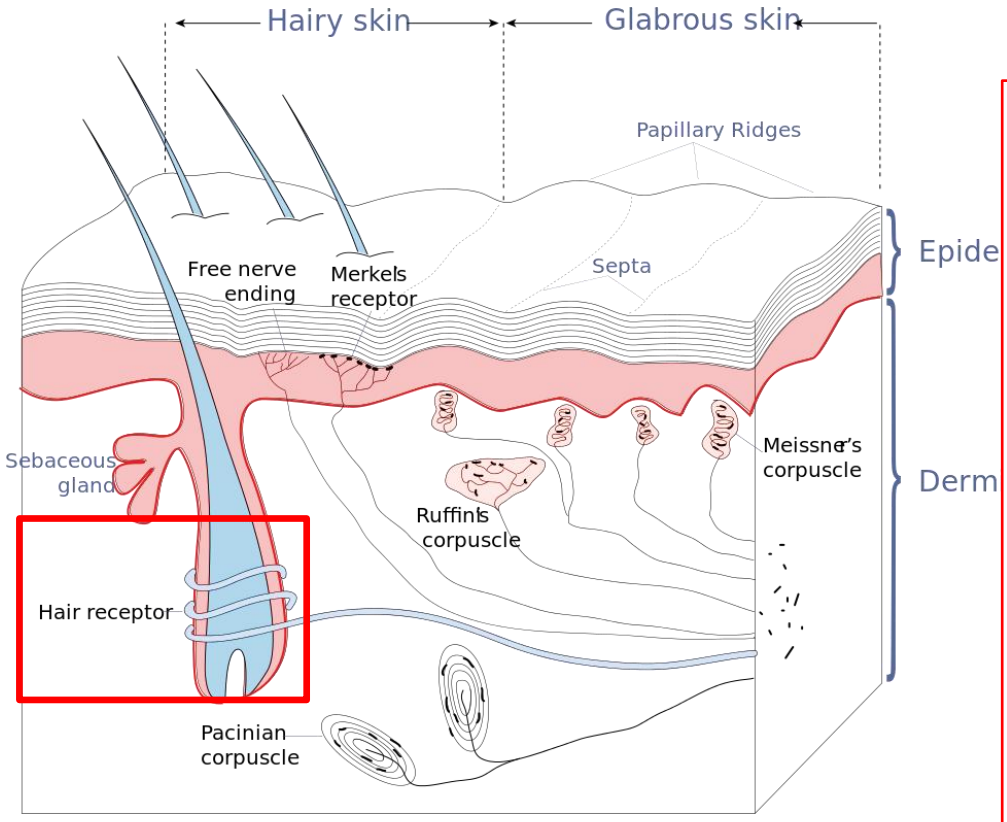
Pacinian corpuscles



- found in deeper layers of dermis
- rapidly adapting
- stronger stimulus – 'deep' touch, poke
- high frequency **vibration**

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Hair follicle receptor

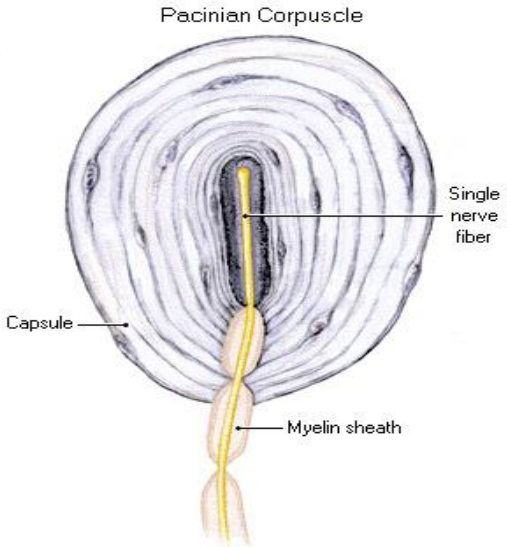


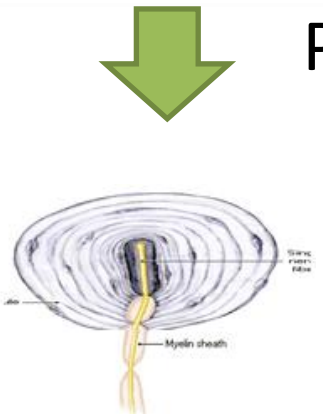
- light touch, but activation in dermis
- rapidly adapting
- constantly changing stimulus required
- nerve fibre wrapped around hair
- hair deflection detected once a stimulus be there

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Signal transduction of skin sensory

- Sensory Transduction: conversion of a sensory stimulus from one form (**mechanical**) to another **(electrical)**
- **Example:**
  - Mechanical movement/force is detected by mechanoreceptors
    → deformations in the skin (dermis and epidermis)
    → trigger the opening of mechanosensitive ion channels (Na+)
    → Depolarization & generate an action potential

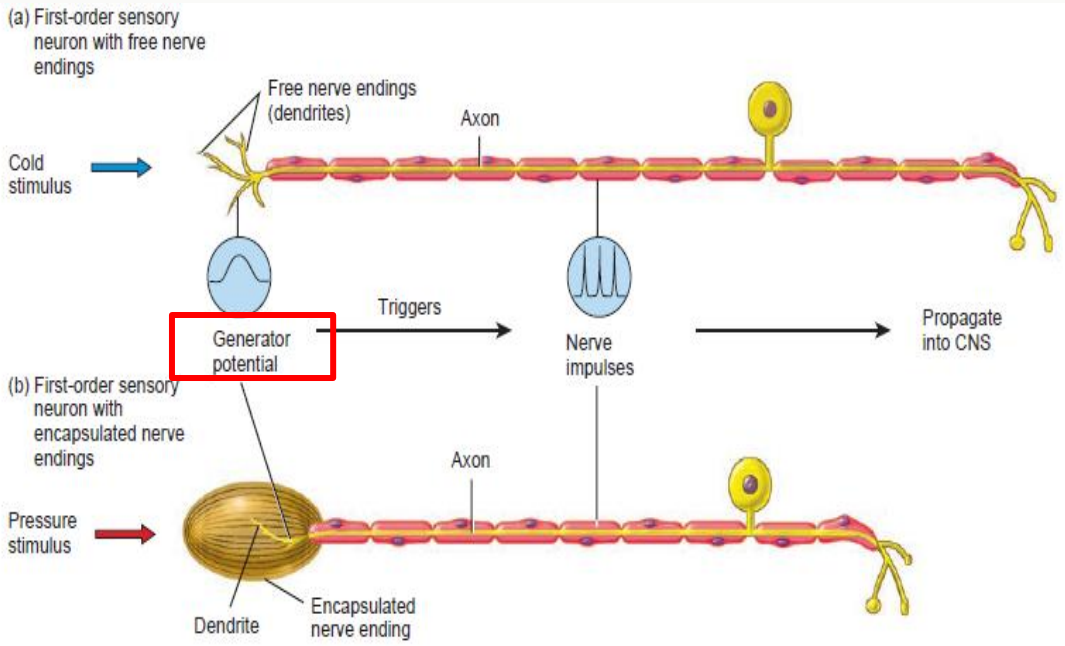


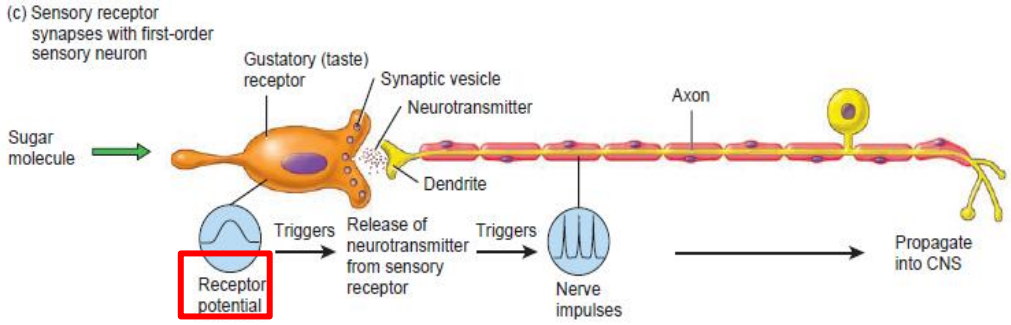


Pressure

Depolarization & action potential generation

Graded Potentials & Receptor types



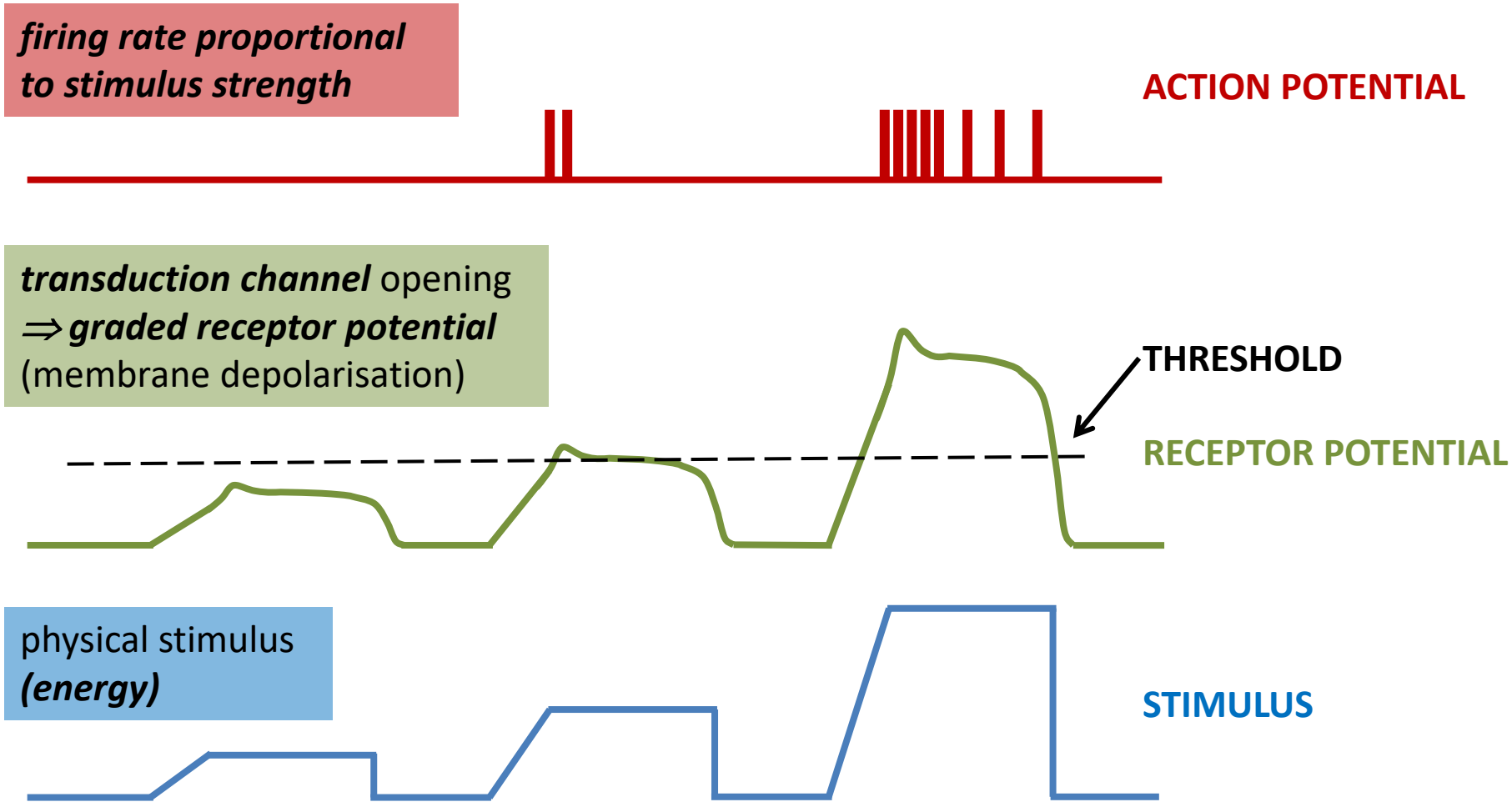


Sensory receptors produce two different kind of Graded potential. Those graded potential eventually propagate (if crosses threshold value) and reaches to brain to elicit a response.

1. Generator potentials are produced mainly by neuronal cells. The dendrites or modified part of neuron acts as a sensory receptor (**Skin receptors mainly**)

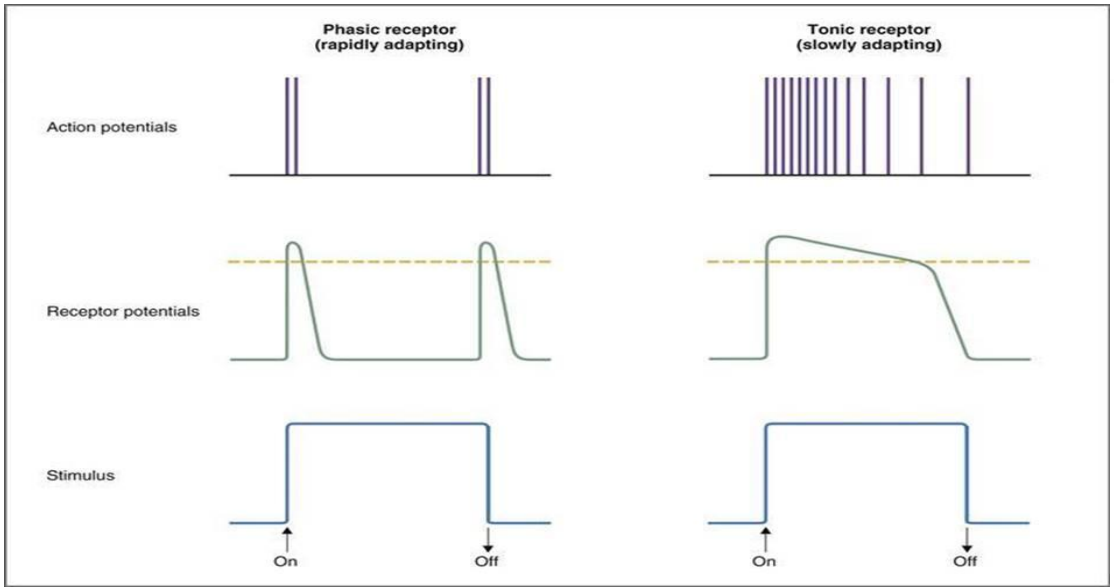
2. Receptor potential are mainly developed by distinct receptor cells eg: gustatory cells, Hair cells of ear, photo receptors of eye. (**Special sense**)

Sensory Receptors - signal transduction



Sensory adaptation

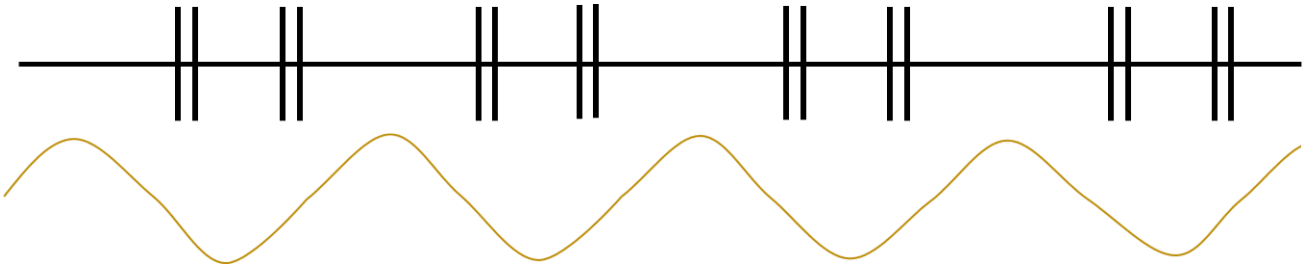
- Adaptation is observed when a constant stimulus is applied for a period
- It reduces the sensitivity towards stimuli and makes us more aware on other stimulus or attention.
- Slowly adapting receptors (tonic receptors) continue responding to a stimulus.
- Rapidly adapting receptors (phasic receptors) respond only at the onset and offset of stimulation.
- Our brain will percept and interpret this accordingly



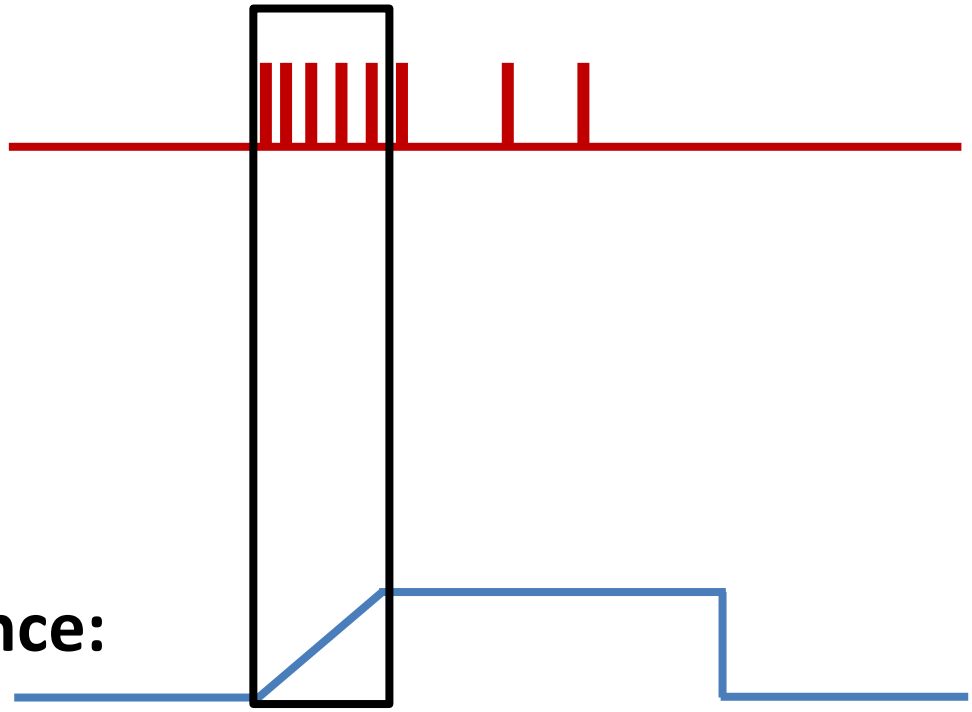
Sensory adaptation

Rapid adapting phasic receptor:

- Examples: Meissner corpuscles and Pacinian corpuscles
- Respond to **application/removal of stimulus**
- Not responding within the maintained and prolonged stimulus



Adaptation of Sensory Responses



fast-adapting or
phasic receptor

Importance:

STIMULUS

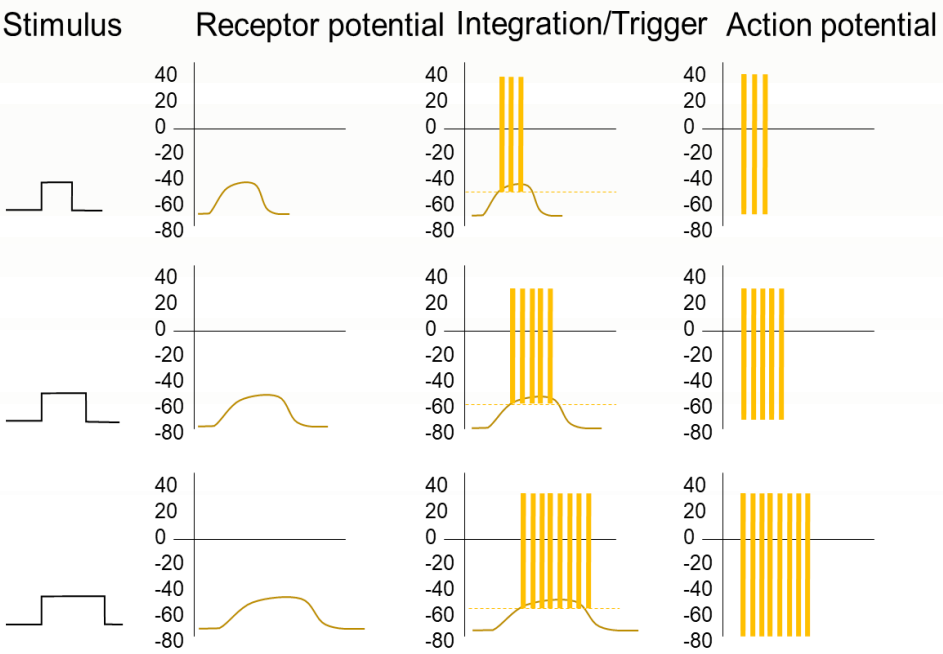
They prevent sensory overload by ignoring constant, unimportant background stimuli, allowing your brain to instantly detect & focus on *new* or *changing* dangers

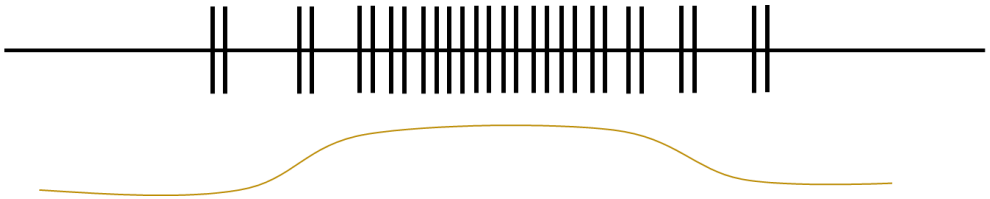
— you immediately feel a mosquito land on your arm
— you stop noticing the constant pressure of your clothes against your skin.

Sensory adaptation

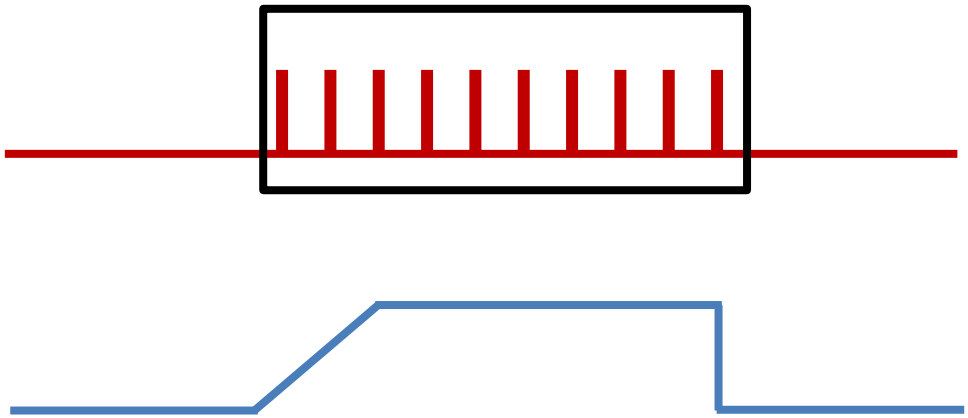
Slow adapting tonic receptors:

- Examples: Ruffini's corpuscles, Merkel's disc and free nerve ending
- Encode stimulus intensity
  - Remains active along the maintained and prolonged stimulus.





Adaptation of Sensory Responses

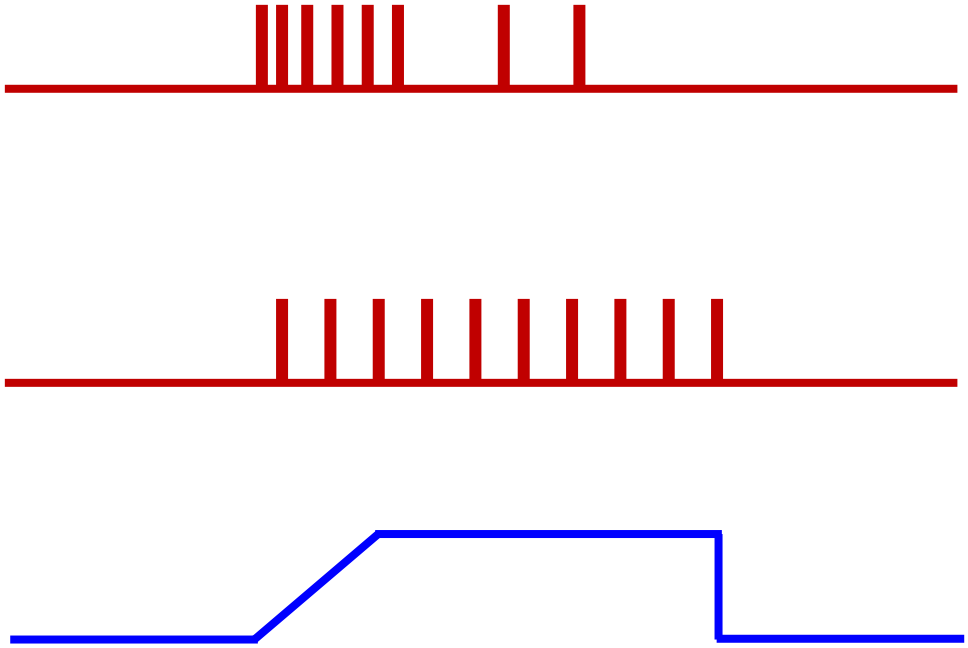


slow-adapting or *tonic* (non-adapting) receptor

STIMULUS

Slow or non-adapting, *important when maintaining information about a stimulus is valuable*
e.g. amount of stretch or pain

Adaptation of Sensory Responses



Phasic receptor
fast-adapting

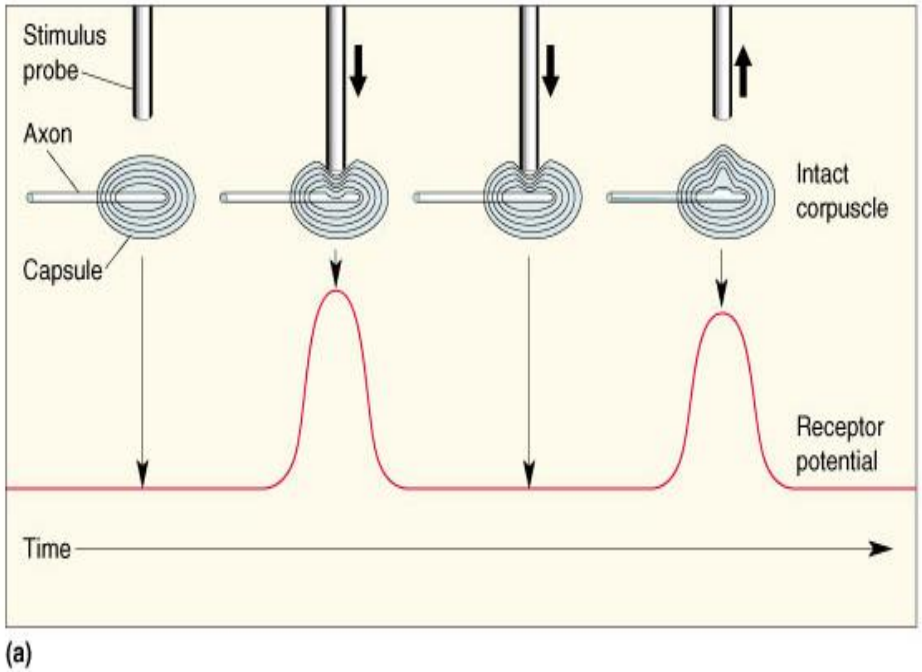
Tonic receptor
slow-adapting or
tonic (non-adapting)

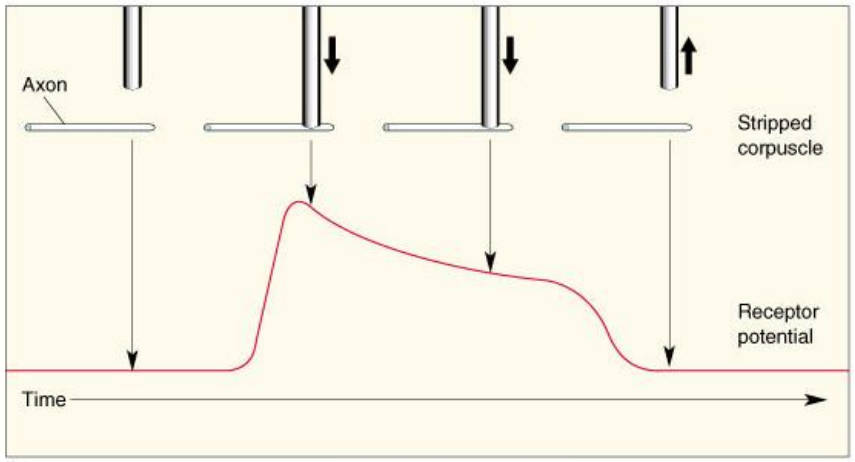
STIMULUS

Slow adaptation detects ***strength of a stimulus***

Fast adaptation detects ***how fast it changes***

Receptor response (e.g.: Pacinian corpuscle) depends on mechanical properties





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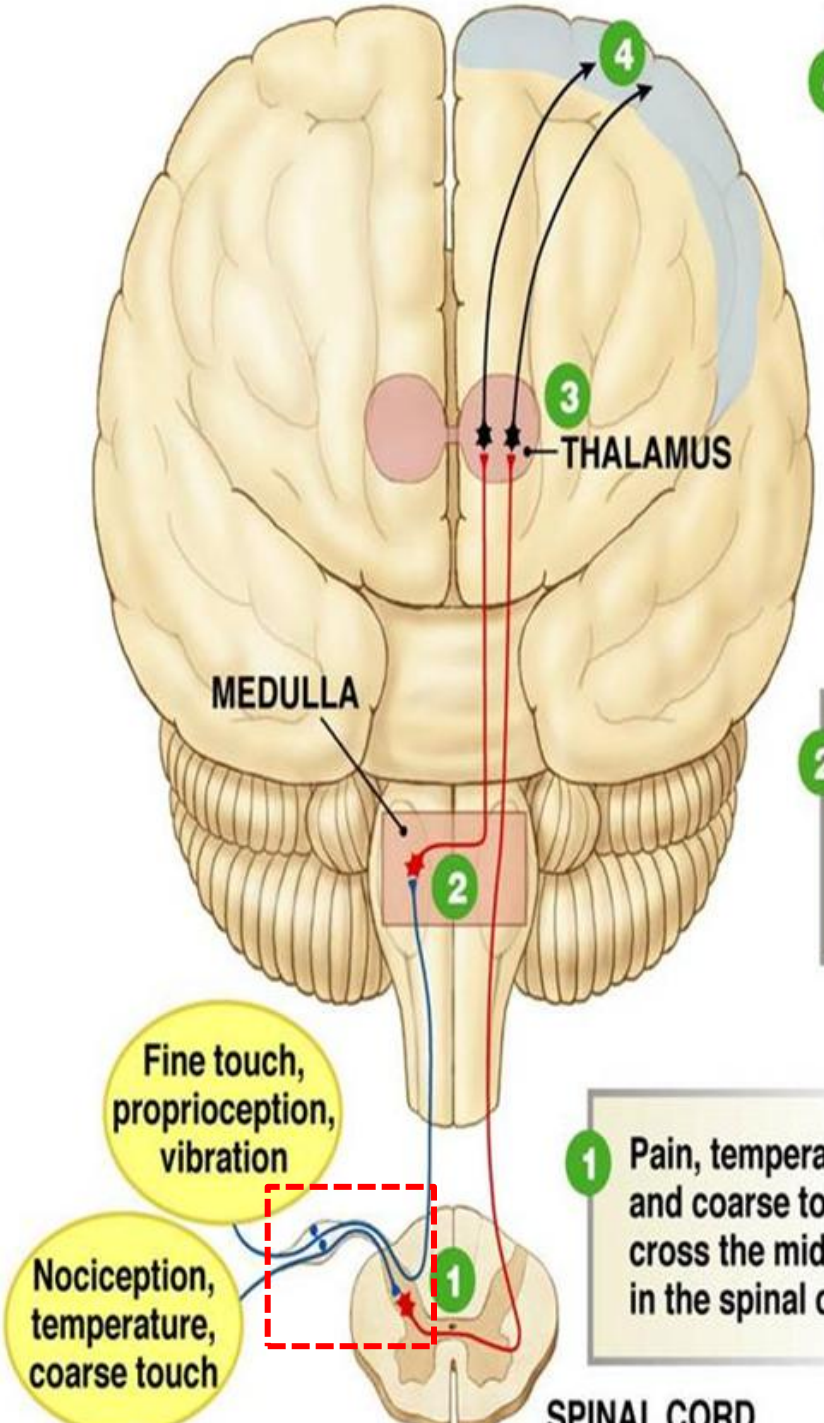
Capsule present -rapid adaptation

Fires mostly at the onset in presence of a stimulus and then stops even though stimulus is present. Comes active only again to an arrival of a repeated new stimulus or at the point of stimulus withdrawal.

Capsule removed- Reduced adaptation

Objectives

- Primary sensory neurons and somatosensory system.
- Different receptors in skin and their properties
- Explain how are stimuli contacting the skin are transduced into neural activity.
- Pathways Carrying sensations of touch, pressure and vibration
- Pathways carrying sensations of pain and temperature to somatosensory cortex.
- Mapping of the somatosensory cortex



4 Sensations are perceived in the primary somatic sensory cortex.

3 Sensory pathways synapse in the thalamus.

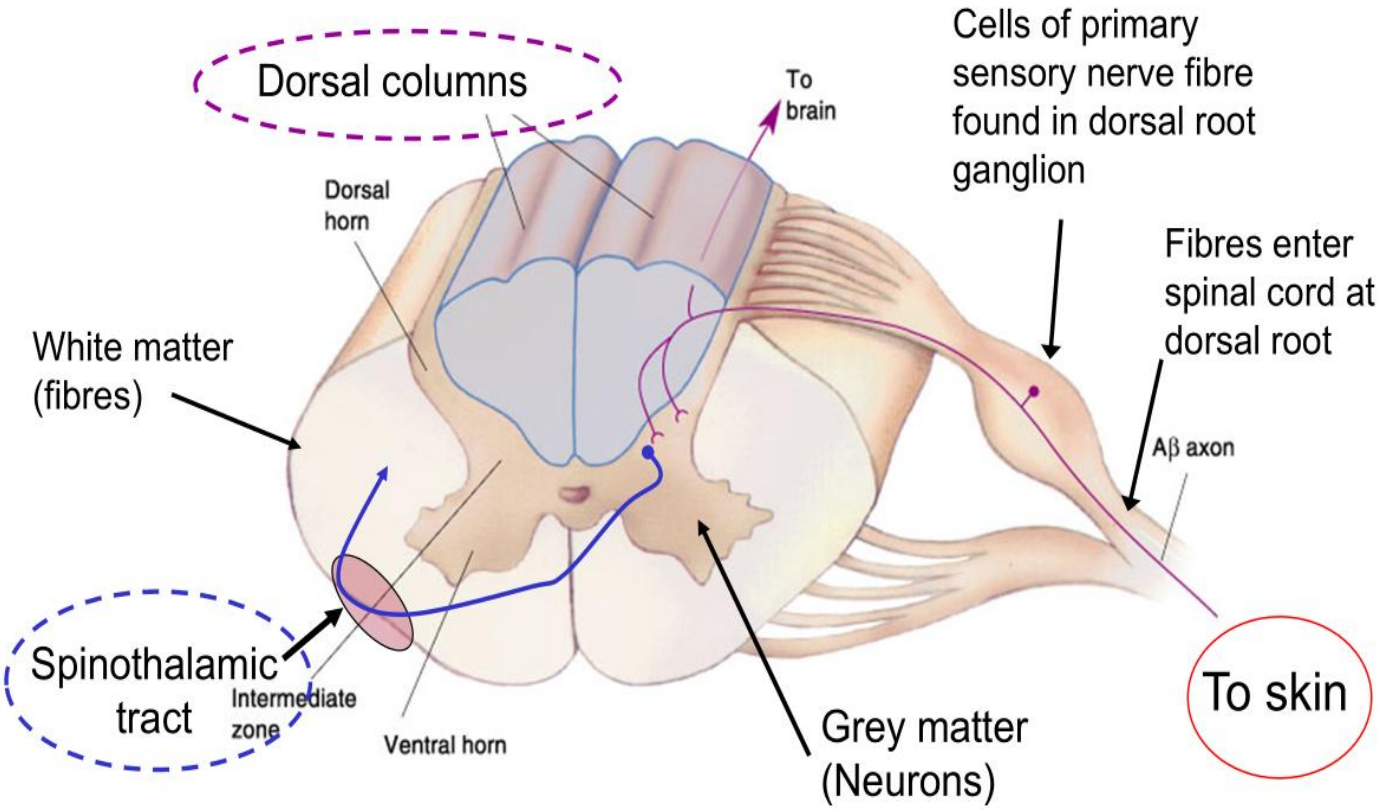
2 Fine touch, vibration, and proprioception pathways cross the midline in the medulla.

1 Pain, temperature, and coarse touch cross the midline in the spinal cord.

KEY

- Primary sensory neuron
- Secondary sensory neuron
- Tertiary neuron

Two pathways for Somatosensation in the spinal cord



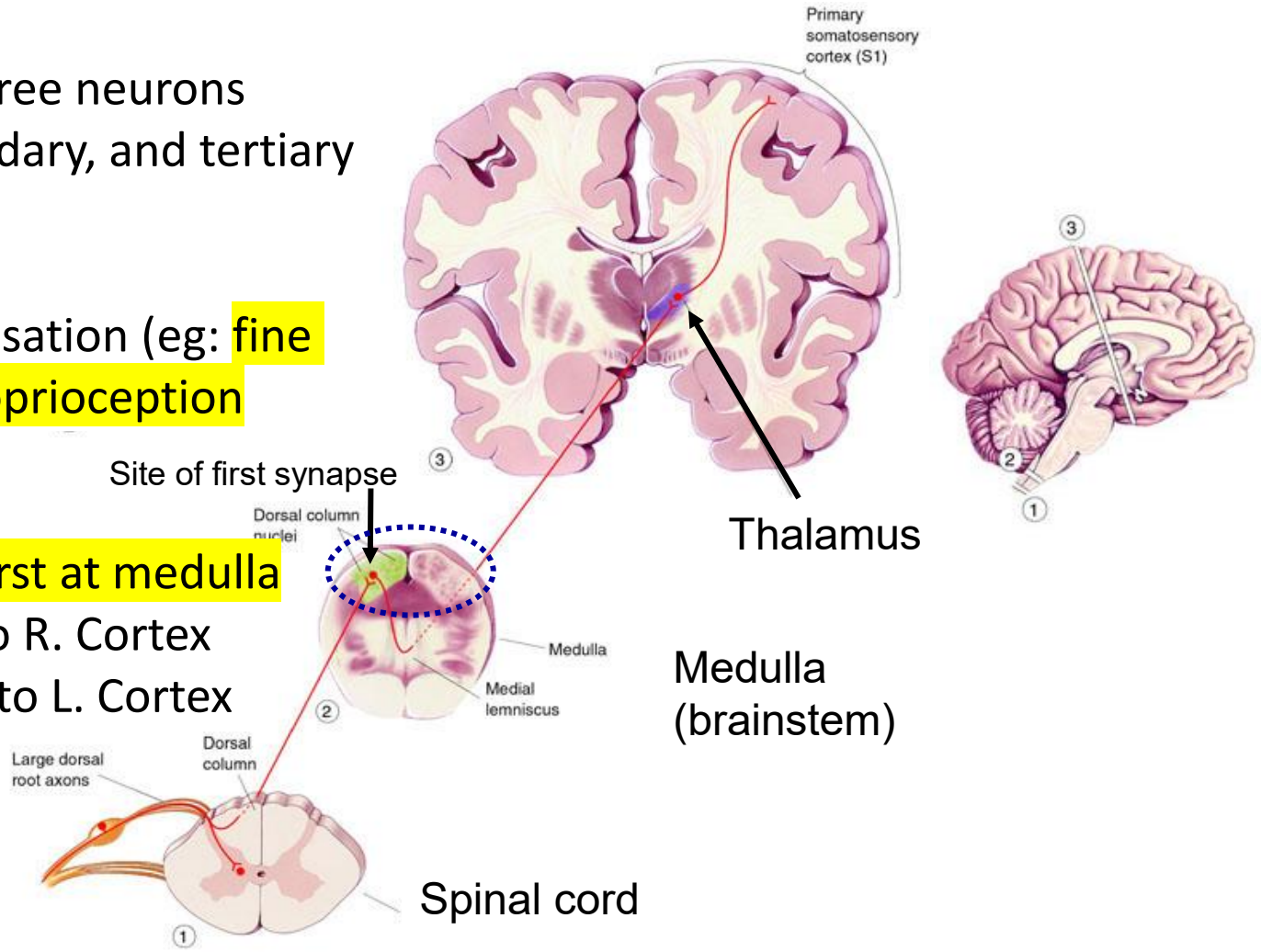
Dorsal column pathway

Typically consist of three neurons types: primary, secondary, and tertiary

Carries information about fine tactile sensation (eg: fine touch), vibration, Proprioception

Information crosses first at medulla

- Left side of body to R. Cortex
- Right side of body to L. Cortex

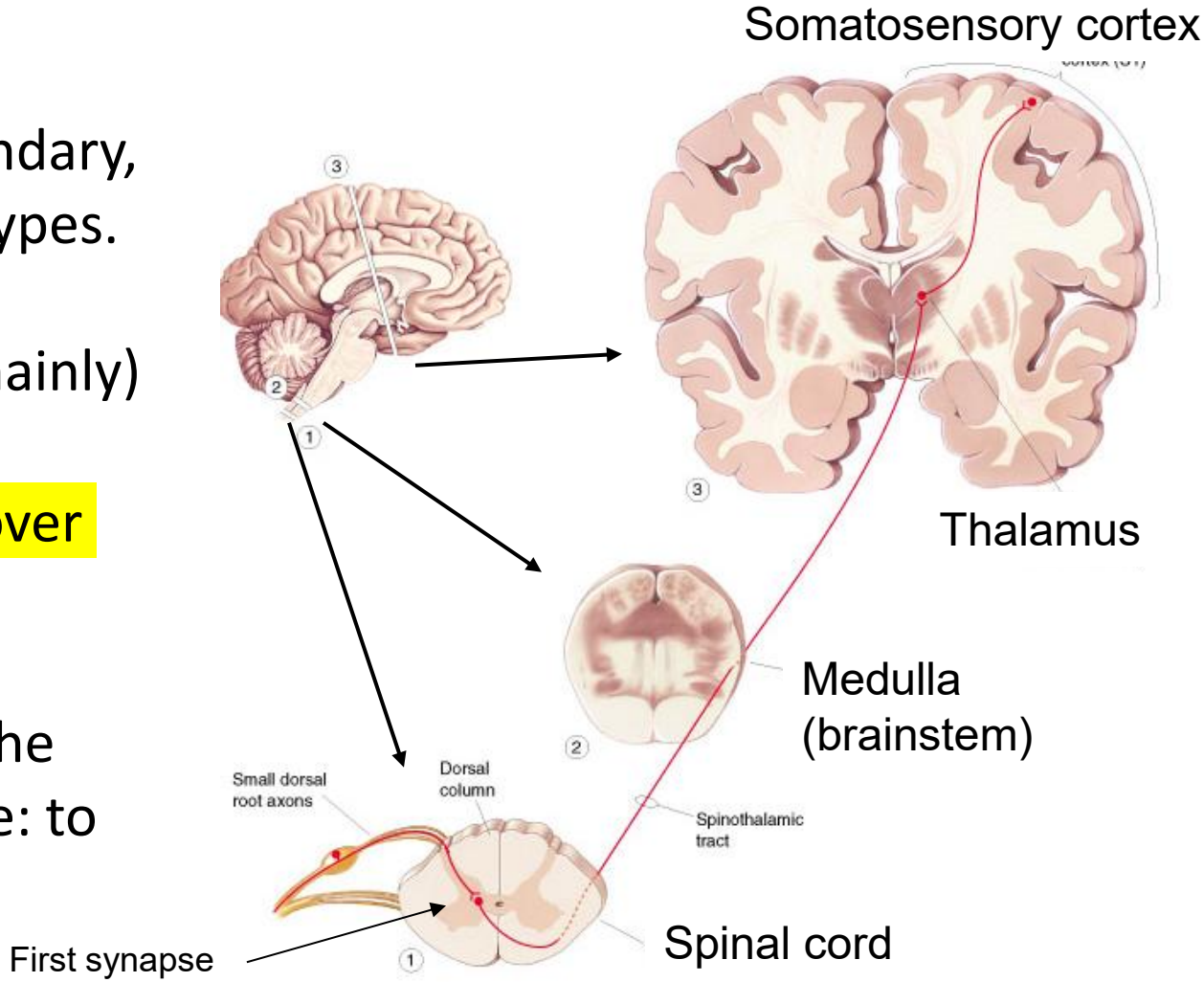


Spinothalamic tract

Having primary, secondary, and tertiary neuron types.
Carries information: Temperature, Pain (mainly)

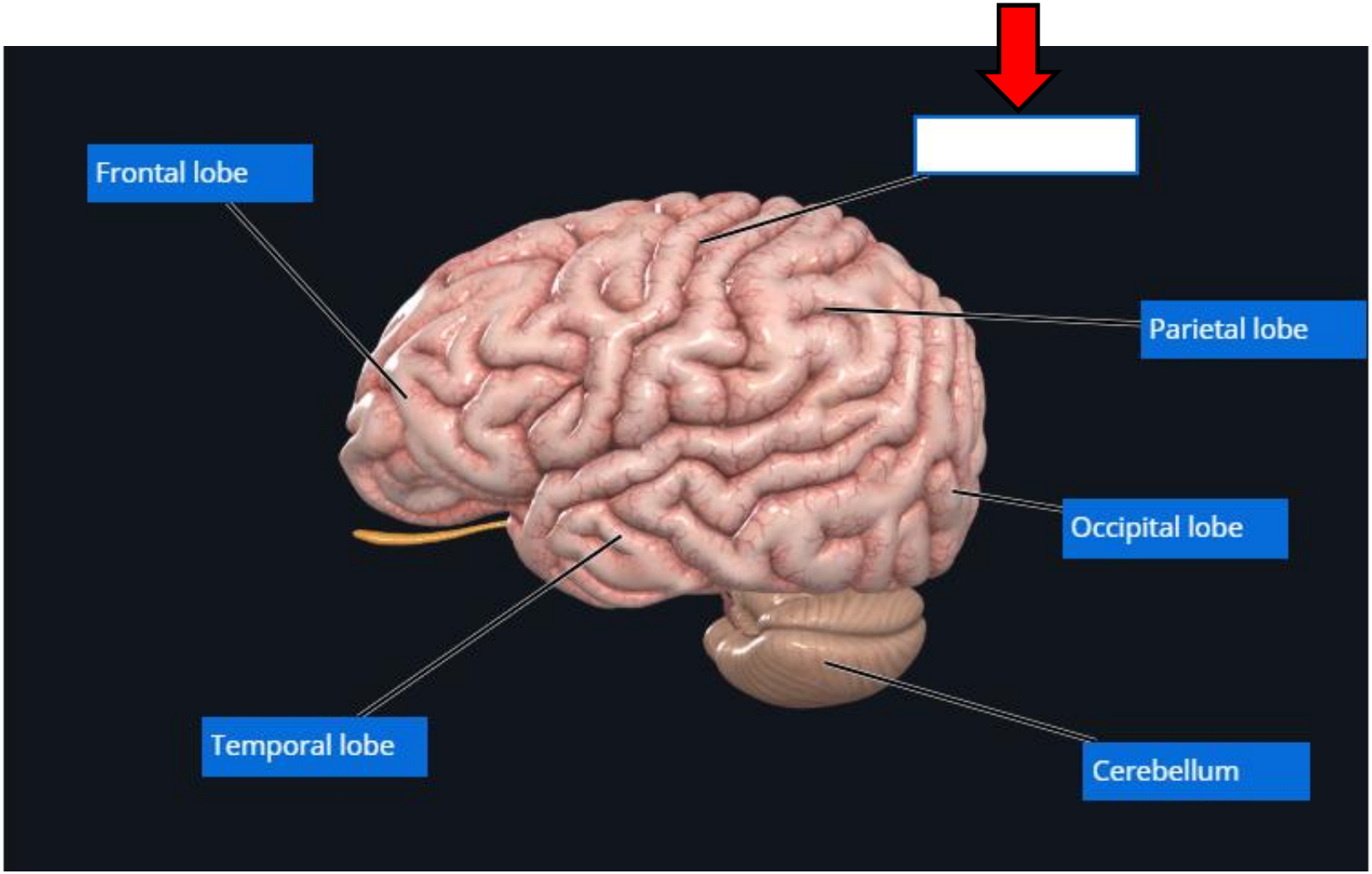
Information crosses over at level of spinal cord

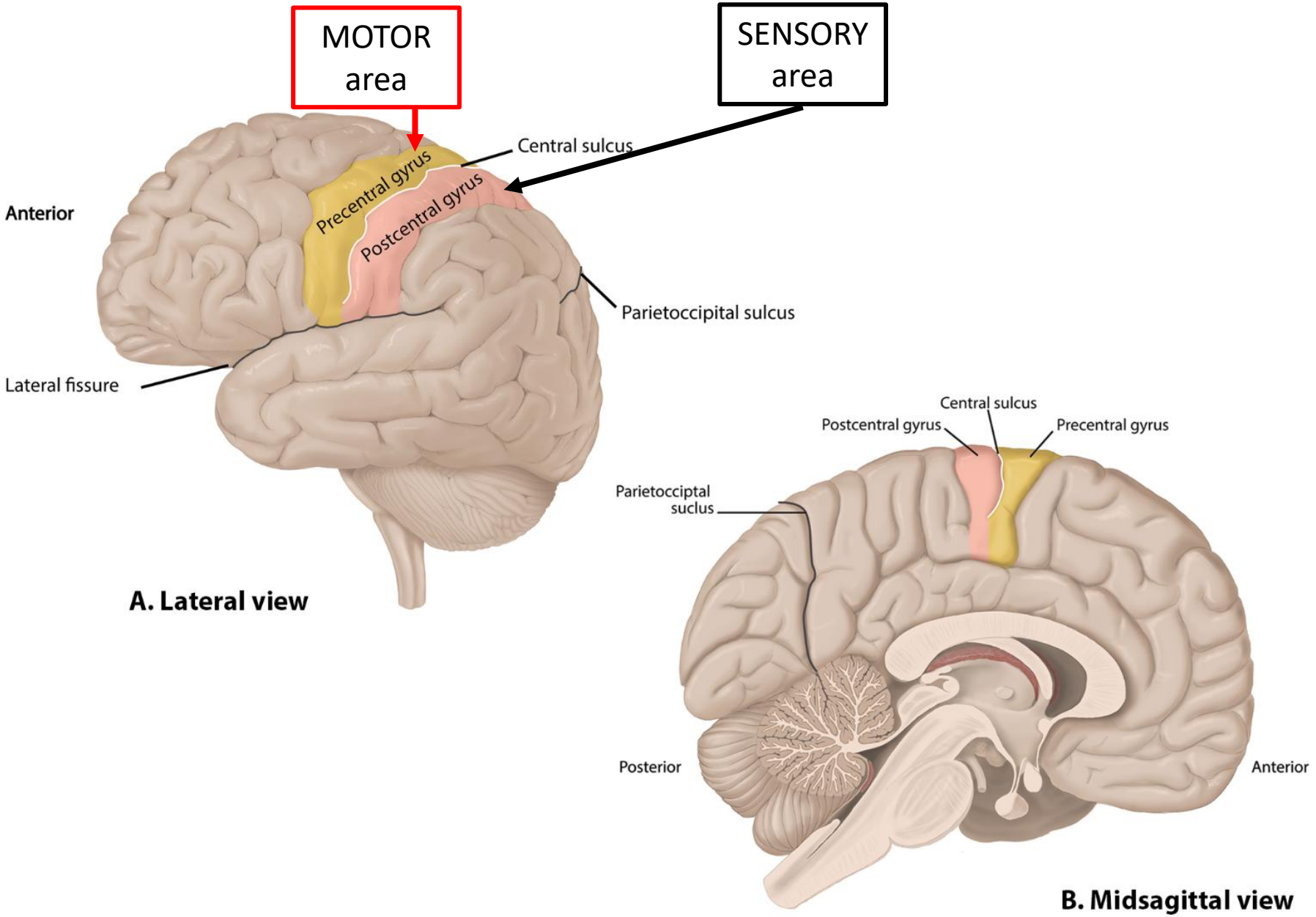
Information goes to the **contralateral brain** (ie: to the opposite side)



Basic topography of brain & Spinal cord (discussed in SGT)

What is this structure?




MOTOR area
SENSORY area
Central sulcus
Precentral gyrus
Postcentral gyrus
Anterior
Parietoccipital sulcus
Lateral fissure
A. Lateral view
Central sulcus
Postcentral gyrus
Precentral gyrus
Parietocciptal suclus
Posterior
Anterior
B. Midsagittal view

Cortical Representation Somatosensation

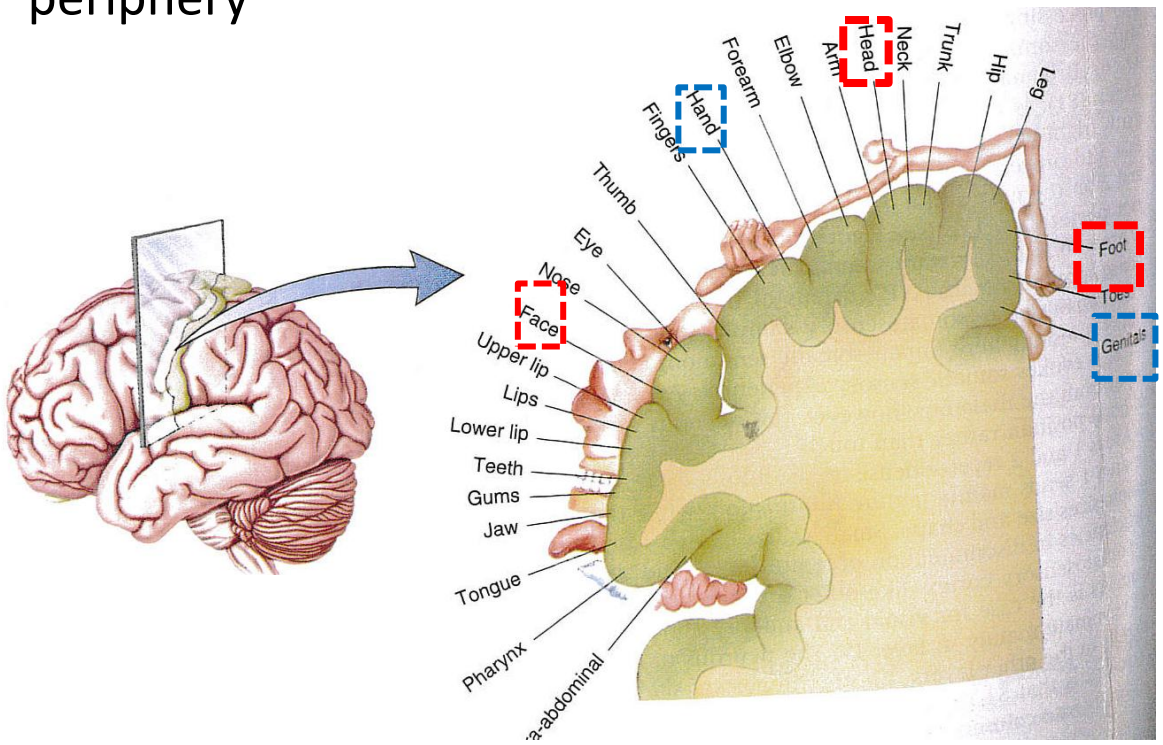
- Somatosensory cortex – post central gyrus (Parietal Lobe)
  - Termination thalamo-cortical pathway (ie: pathway from thalamus to cortex). Thalamus receives huge amount of sensory information from periphery.

- Somatosensory cortex contains a topographical map of senses
  - Proportionate to amount of sensory information
  - Tongue, lips, fingertips – large area
  - Arms, legs – relatively small area

- Brodman's areas 1,2,3

Organization in Somato Sensory Cortex (SSC)

Mapping of the body's surface sensations happens onto a structure of brain based on sensory information received from periphery

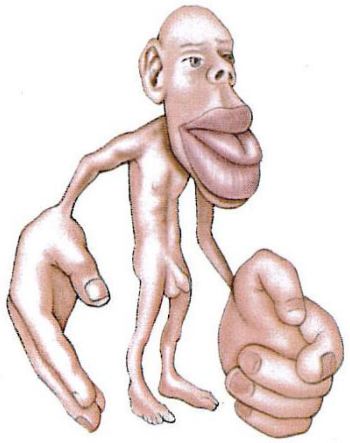


Features:

1. **Not always continuous**

2. **Not Scaled properly**

(Eg: Fingertips, lips, hands provided bigger space. Dependent how much sensory input brain receives from periphery)



Summary:

- The somatosensory system transmits information from sensory receptor organs in skin, muscles, joints and viscera to the cerebral cortex where conscious awareness happens
- Sensory receptor produce 2 types of graded potential
- Adaptation is an important property and get modified with structural alteration
- Small RF provides precise perception compared to big ones.
- Temperature & pain sensation is carried by Spinothalamic tract whereas dorsal column pathway is responsible for fine touch, pressure, proprioception
- Somatosensory cortex receives the dense input from thalamus and lesions produce significant impact on perceiving tactile sensations.
- Scaling is dependent on amount of sensory input received from periphery

Important Terminologies :

1. **Sensory Transduction:** The process of converting a sensory stimulus (like mechanical pressure or temperature) from the external environment into an electrical signal (a graded potential or action potential) that the nervous system can understand. For example, when you press on the skin, mechanoreceptors convert that mechanical force into an electrical signal.

2. **Sensory Adaptation:** A decrease in the response of a sensory receptor to a constant, unchanging stimulus over time. This property helps the brain filter out unimportant, background information so it can focus on new or changing stimuli. Two main types: Phasic (Rapidly Adapting) Receptors **and** Tonic (Slowly Adapting) Receptors.

3. **Graded Potential:** A small, local change in the membrane potential of a cell (like a neuron or receptor cell). Its size (amplitude) is **"graded"**—meaning it varies directly with the strength of the incoming stimulus. A stronger stimulus produces a larger graded potential, while a weaker one produces a smaller one. It does not follow the "**All or None Law**" and it is "**local and decremental**".

Important Terminologies :

5. Dorsal Column Pathway: A major ascending pathway in the spinal cord that carries information for fine tactile sensations (fine touch), vibration, and proprioception (body position) to the brain. Information in this pathway crosses over to the opposite side of the brain at the level of the medulla (brainstem).

6. Spinothalamic Tract: The primary ascending pathway in the spinal cord that carries information about pain and temperature to the brain. Unlike the dorsal column pathway, the information in this tract crosses over to the opposite side of the spinal cord at the level it enters.

7. Somatotopic Organization (and the Homunculus): The topographical mapping of the body's surface onto specific areas of the somatosensory cortex (postcentral gyrus). This map is not to scale; body parts with a high density of sensory receptors and greater sensory importance (like the lips and fingertips) are represented by much larger areas of the brain than areas like the trunk or legs. This distorted, upside-down "map" is often called the sensory **homunculus** ("little man").

Lets have some questions to solve after the session



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